

Difference between realtime operating systems and realtime embedded systems.

Feature 	Real-Time Operating System (RTOS)	Real-Time Embedded System
Nature	Software (Operating System Kernel)	System (Hardware + Software)
Primary Goal	Deterministic scheduling & low latency	Specific, time-constrained functionality
Resource Usage	Minimal, low overhead	Varies, but designed for resource-constrained environments
Scope	Manages tasks/interrupts	Controls a specific device (e.g., sensor, motor)
Components	Scheduler, tasks, IPC, drivers	Processor, RAM/ROM, sensors, actuators